

Yahya Zakaria

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Education

Bachelor of Computing and Data Science, Alexandria University

Sep 2023 – Jul 2027

- CGPA: 3.9839/4.0

Experience

Software Engineer Intern, Deloitte Innovation Hub – Cairo, Egypt

Jul 2026 – Aug 2026

- Designed and implemented an internal **CV search system** using **Spring Boot**, **Python**, **RabbitMQ**, **PostgreSQL**, and **Vertex AI**, processing batches of **50 CVs**.
- Built the **Dockerized** environment and **CI/CD pipelines**, and resolved critical issues across the end-to-end processing pipeline.
- Collaborated within a Scrum team to design, implement, test, and deliver the system within **4 weeks**.

Coding Instructor, Pioneer Academy – Egypt (Remote)

Feb 2026 – May 2026

- Explained courses to college students, such as Regression, OOP, and problem-solving.

Research Intern, Nile University - Cairo, Egypt (Hybrid)

Jun 2025 – Sep 2025

- Conducted data preprocessing, EDA, and geospatial data analysis on the NetMob 2025 dataset (~**500 million rows**) using PySpark and Kepler.gl, Uncovered the most optimal locations for the installation of piezoelectric materials in Île-de-France.
- The research paper was presented as a poster at the **NetMob 2025 conference in Paris**.

Big Data trainee, Samsung Innovation Campus - Cairo, Egypt (Hybrid)

Jul 2024 – Sep 2024

- Learned SQL, Hadoop ecosystem, Cassandra, MongoDB, Spark, and more about Big Data and Data Visualization.

Projects

PavementEye | Road-Cracks detection System

[GitHub Repo](#)

- Fine-tuned a computer vision model (YOLOv8s) on the **EGY_PDD research dataset**. Developed a robust **streaming data pipeline** using Apache Kafka, Spark, Cassandra, and Docker, and deployed the system on Azure. Integrated detections with **OpenStreetMap** to enrich **geospatial analysis**.
- Processed **1 image/second** and detected **100+** pavement cracks across **11 crack types** in **under 30 minutes**.
- Top **50 finalists** in the Huawei Developer Competition 2025 – Northern Africa.

Automated OT/ICS Threat Intelligence Agent

[GitHub Repo](#)

- Built an automated pipeline to extract OT/ICS CVEs from the NVD API every 10 minutes using **two-stage filtering**: heuristic keyword matching followed by reasoning with a local **LLM (Llama 3.2)**, generating actionable risk insights.
- Classified **69 OT/ICS CVEs** and developed a Streamlit dashboard with impact/exploitability visualizations and advanced filtering.

Exam Timetabling for Buraydah Colleges | Graph Coloring

[GitHub Repo](#)

- Worked on real data from Buraydah Colleges to solve the exam scheduling problem using **graph coloring**.
- Successfully distributed **1820 Students** across the specified room capacities (**18 rooms** with **2190 seats**), **6-time slots**, and **496 registered courses** with no conflicts in the minimum time needed (**8 days**) after optimizing time with resources.

Skills

Technical Skills:

Languages: Python, Java, C++ , JavaScript, SQL, Bash.

AI & Data: Hadoop, NumPy, Pandas, GeoPandas, Matplotlib, Seaborn, Plotly, Scikit-learn, PySpark, SSIS, Apache Kafka, Power BI, Apache Airflow, Apache NiFi.

Other Tools: Kepler.gl, Docker, Git & GitHub, LaTeX.

Software fundamentals: OOP, Data Structures & Algorithms, Databases(SQL & NoSQL), Computer Networks, Cloud Computing, Operating Systems.

Soft Skills: Problem Solving, Team Leadership, Planning, Research Skills.

Achievements and Certifications

- SprintUp – AI and Machine Learning Foundations, Sprints 2025
- Ranked **5th** within faculty, **25th** overall on contest day, ECPC 2024
- Python for Data Analysis, NTI 2024

Volunteer work

Manager of Projects, IEEE ALEX CSC

Jul 2026 – Present

- Built an automated coding challenge platform using **Go**, **PostgreSQL**, and **GitHub Actions CI/CD pipelines**.

Problem Solving Trainer, ICPC FCDS

Dec 2024 – Jun 2025

- Facilitated **275 Students** in competitive programming by creating **5 video explanations** for problems and answering their questions.
- Created **6 problems** on polygons using LaTeX for contests and sheets on Codeforces.
- Explained sets, maps, and priority queues in C++ to students in an offline session.